

Professional Summary

- AI/ML Engineer with 4.5+ years of experience in designing, developing, and deploying scalable machine learning solutions across NLP, computer vision, forecasting, and generative applications, delivering measurable impact in production environments.
- Expert in creating full-stack ML pipelines using Python, Scikit-learn, Pandas, MLflow, and XGBoost; experienced in data cleaning, feature selection, model tuning, and containerized deployments with Docker, FastAPI, and AWS SageMaker.
- Skilled in building and fine-tuning deep learning models using PyTorch, TensorFlow, and Keras for OCR, image recognition, time series analysis, and large-scale classification; optimized for inference speed and GPU memory efficiency.
- Proficient in applying transformer architectures such as GPT-4, BERT, LLaMA, and T5 for NLP tasks like semantic search, document summarization, text classification, and Retrieval-Augmented Generation (RAG) pipelines using LangChain and Hugging Face.
- Strong knowledge of Generative AI tools including OpenAI API, Stable Diffusion, DALL-E, and prompt engineering techniques; coupled with advanced understanding of MLOps, version control (DVC), experiment tracking, SQL, and visualization using Power BI and Tableau.

Education

Master of Science in Data Science (GPA: 3.7/4) | University of New Haven, West Haven, Connecticut,

Skills

Language/ IDE's: Python, MATLAB, Jupyter Notebook, Google Colab, VS Code, SSMS

Packages and Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, TensorFlow, Keras, NLTK, XGBoost, PyTorch

Machine Learning: Linear, Logistic Regression, Decision Trees, Random Forests, A/B Testing, Naive Bayes

Deep Learning: CNN, RNN, LSTM, NLP, Large Language Model (LLM), LangChain, Hugging Face Transformers (BERT, GPT-3)

Cloud/Visualizations: AWS (EC2, SQS, SNS, Code Deploy, CloudWatch, API Gateway), GCP (Vertex AI, Google Cloud Storage), Tableau, Power BI

Statistical Techniques: Hypothesis Testing, Data Visualization, Data Modelling, A/B testing, Model Evaluation

Database and Tools: SQL Server, MySQL, PostgreSQL, Redis

Certifications: [Python Programming Course](#), [DevOps for Data Scientist](#), [Amazon Web Services \(AWS\) Certified](#), [Data Analysis with Python](#)

Work Experience

CVS Health, Hartford, Connecticut

May 2024 - Present

Machine Learning Engineer

- Developed Generative AI models using fine-tuned BERT and transformer architectures to auto-summarize unstructured clinical notes and transcripts, reducing documentation time by 35% for providers and improving EHR data consistency across MinuteClinic locations.
- Built chronic disease risk prediction models using XGBoost and patient vitals/history data from Aetna claims, identifying high-risk individuals with 89% accuracy and enabling early intervention programs for diabetes and hypertension management.
- Integrated NLP pipelines using spaCy and rule-based matchers to extract and classify medical terms, procedure codes, and medications from prior authorization forms, automating a manual process and improving workflow efficiency by 50%.
- Applied anomaly detection on pharmacy transaction data using Isolation Forest and statistical modeling to flag unusual refill patterns, contributing to a 22% improvement in fraud detection and opioid monitoring compliance.
- Collaborated with cross-functional teams to fine-tune domain-specific LLMs for internal use cases like member service automation and document classification, improving response accuracy and cutting resolution time for claim-related queries.
- Utilized Pandas and NumPy to build efficient data preprocessing pipelines for handling multi-source datasets, reducing data wrangling time by 40% and ensuring consistent feature formatting for downstream ML model training and evaluation.
- Constructed retrieval-augmented generation (RAG) solutions using LangChain to connect LLMs with CVS internal repositories and medical references, streamlining the preparation of case summaries and authorization documents.

Hexaware Technologies, India

Aug 2021- Dec 2022

AI/ML Engineer

- Successfully deployed multiple machine learning models on AWS EC2 by building automated retraining workflows, setting up monitoring tools, and integrating CI/CD pipelines. This led to a 30% improvement in deployment stability across production environments.
- Refined a BERT-based sentiment analysis model on over 20,000 customer feedback entries. The project boosted classification accuracy by 22% and equipped the UX and product teams with key emotional insights for better decision-making.
- Embedded OpenAI GPT-3 with prompt engineering and few-shot learning to generate real-time summaries of lengthy customer logs. This initiative reduced average response time by 41% and significantly enhanced customer support performance.
- Engineered a scalable document classification pipeline combining OCR technologies with TensorFlow CNNs. This system processed more than 9,000 scanned documents monthly and reduced compliance-related turnaround time by 50%.
- Created custom NLP pipelines using Hugging Face Transformers and NLTK for entity recognition and keyword extraction, enabling faster analysis of qualitative survey data and helping marketing teams identify emerging consumer trends 18% quicker.
- Designed and maintained interactive Tableau dashboards to present NLP outputs such as sentiment drift, topic clusters, and model confidence levels. These dashboards provided business stakeholders with actionable insights at a glance.

Hexaware Technologies, India

Jan 2020 - Jul 2021

Junior AI/ML Engineer

- Architected and deployed deep learning models to extract structured data from over 7,000 handwritten patient forms and records monthly, using CNNs and attention-based layers, reducing manual entry work by 65% and improving documentation compliance.
- Devised LSTM and OpenCV-based surveillance models for monitoring ICU and elderly care units, enabling automatic detection of fall risk or abnormal movements, and helping reduce manual video review time by over 55% for hospital monitoring staff.
- Formed predictive models using Random Forest and Logistic Regression to identify patients at risk of missing follow-up appointments or dropping off chronic care programs, achieving 87% accuracy and driving targeted engagement campaigns with care managers.
- Streamlined the training and deployment pipeline for diagnostic and clinical text classification models by automating hyperparameter tuning in TensorFlow and PyTorch, reducing training cycles by 60% and improving model handoff speed across ML and engineering teams.
- Formulated operational dashboards using Power BI to visualize model accuracy, prediction latency, and drift trends in production healthcare AI systems, helping clinical informatics teams maintain transparency and ensure consistent model performance.
- Fine-tuned EfficientNet-based image classification models for automatic tagging of lab forms, prescription scans, and diagnostic images, which improved classification accuracy by 31% and supported better workflow automation in electronic health record (EHR) systems.